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Authors: Mark Kushner

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Block 13: Supplementary Note

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ESCAMPIG Workshop:
**PREDICTIVE AND PRACTICAL
SIMULATIONS OF PLASMA SYSTEMS AND
PLASMA PROCESSES***

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ESCAMPIG XXVI
Brno, Czech Republic
July 2024

* Work supported by US Department of Energy (SC0020232), US National Science Foundation (2020010, 2032604, 2009219) and US Army Research Office (W911NF-18-1-0240, W911NF-20-1-0105).

HOW TO MAKE IMPACT BY MODELING?



$$\frac{\partial f(\vec{v}, \vec{r})}{\partial t} = -\vec{v} \cdot \nabla_x f(\vec{v}, \vec{r}) - \frac{q\vec{E}}{m_e} \cdot \nabla_v f(\vec{v}, \vec{r}) + \left(\frac{\partial f(\vec{v}, \vec{r})}{\partial t} \right)_c$$

**Plasma Equipment
Design**

**Fundamental
Phenomena**

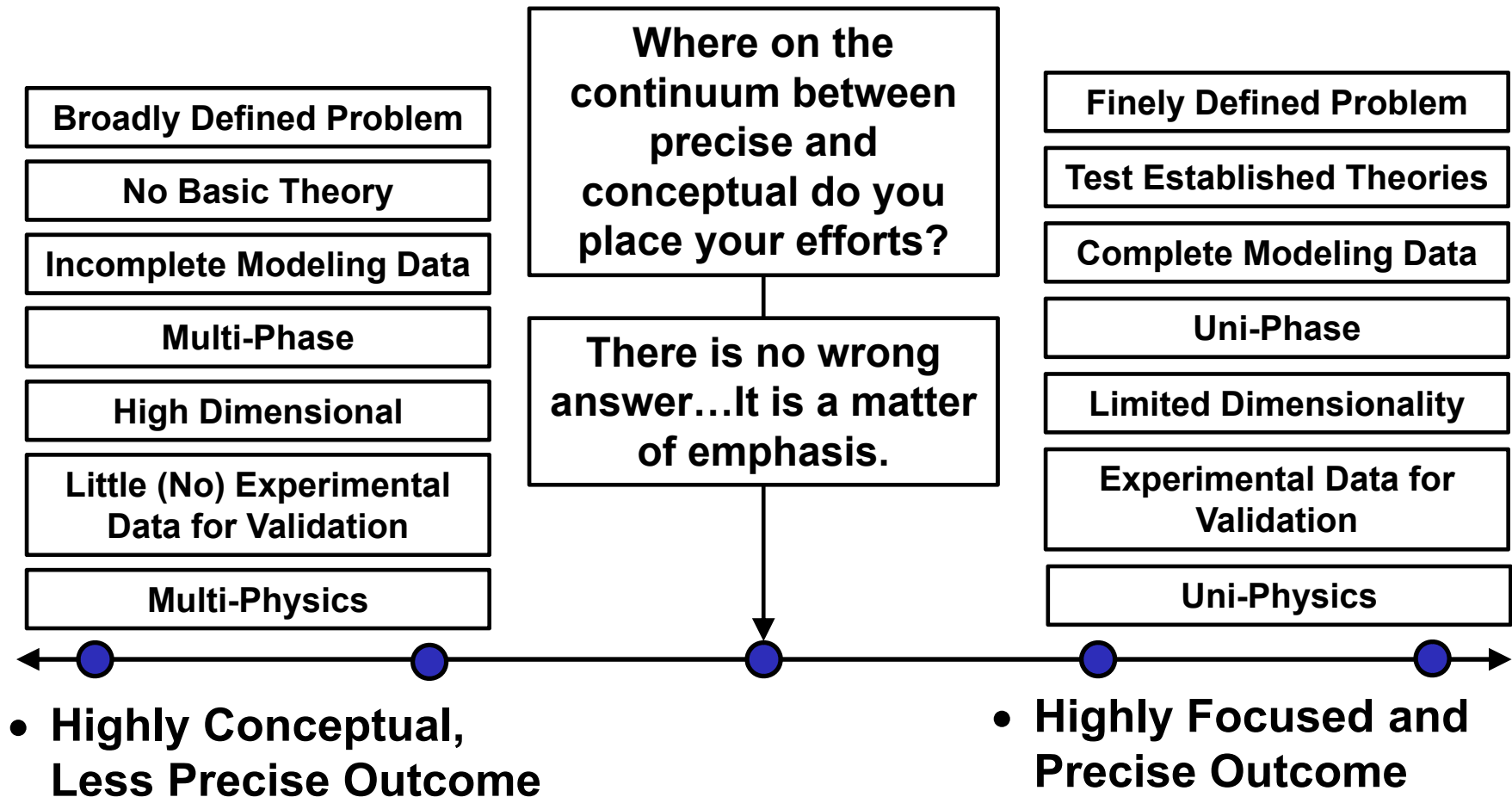
- Low temperature plasma (LTP) modeling spans from fundamental investigations of transport phenomena to design of plasma equipment making *impact from basic science to applications*.
- Models are often developed with one of these goals in mind.
- To do the job correctly, models addressing applications should also properly address the fundamentals.

- <https://www.lamresearch.com/products/our-processes/etch/>

**University of Michigan
Institute for Plasma Science & Engr.**

MODELING FOCUS: PRECISION vs CONCEPTUAL

- In the absence of unlimited computing resources and data, decisions are usually made on “precise” vs “conceptual” outcomes of models.



COMPUTATIONAL EXPLORATION

- **Student:**

“I don’t know the sticking coefficient of Cl on Si, and so I can’t do the simulation”

- **Advisor:**

“Sure you can. You know the value of the sticking coefficient. It is more than zero and less than 1.”

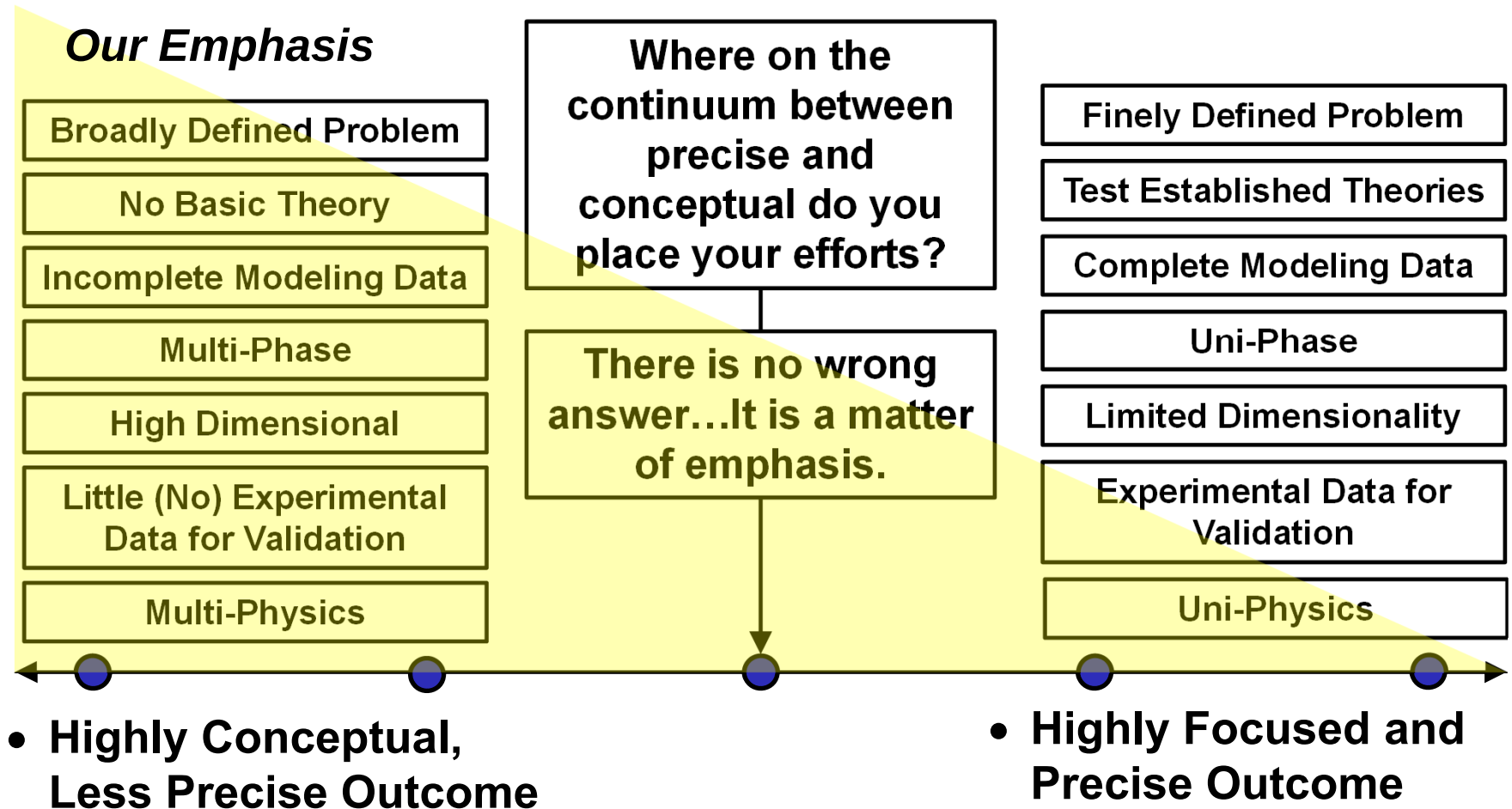
*Use models to investigate unexplored phenomena that are not well characterized due to lack of data.
Refine your conclusions as that data becomes more available.*

UM MODELING GOALS

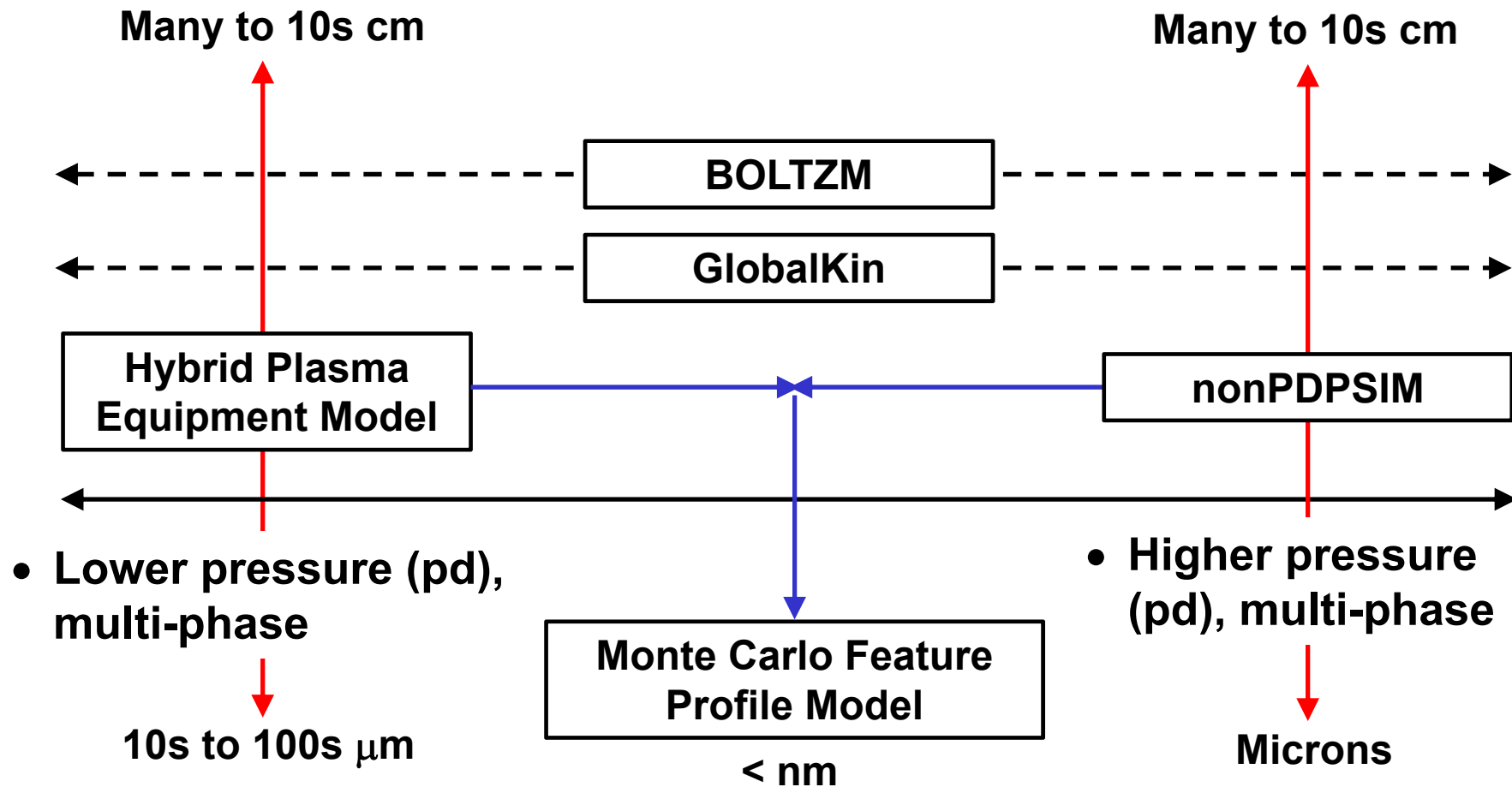
- *Investigate fundamental and applied phenomena in low temperature plasmas and plasma surface interactions through computer simulation capable of equipment design.*
- **To meet goals, create a computational infrastructure that**
 - **Accepts “new knowledge” as that knowledge becomes available (e.g., cross sections) to become more precise.**
 - **Compensates for missing data by using scaling laws and parametric studies.**
 - **Makes a minimum of limiting assumptions.**
 - **Links needed physics (“multi-physics”) and phases.**
 - **Addresses fundamentals while being technology focused.**
 - **Is accessible to users (while not being a commercial product).**
 - **Outlives a single generation of graduate students.**
 - **Leverages other codes.**

MODELING FOCUS: PRECISION vs CONCEPTUAL

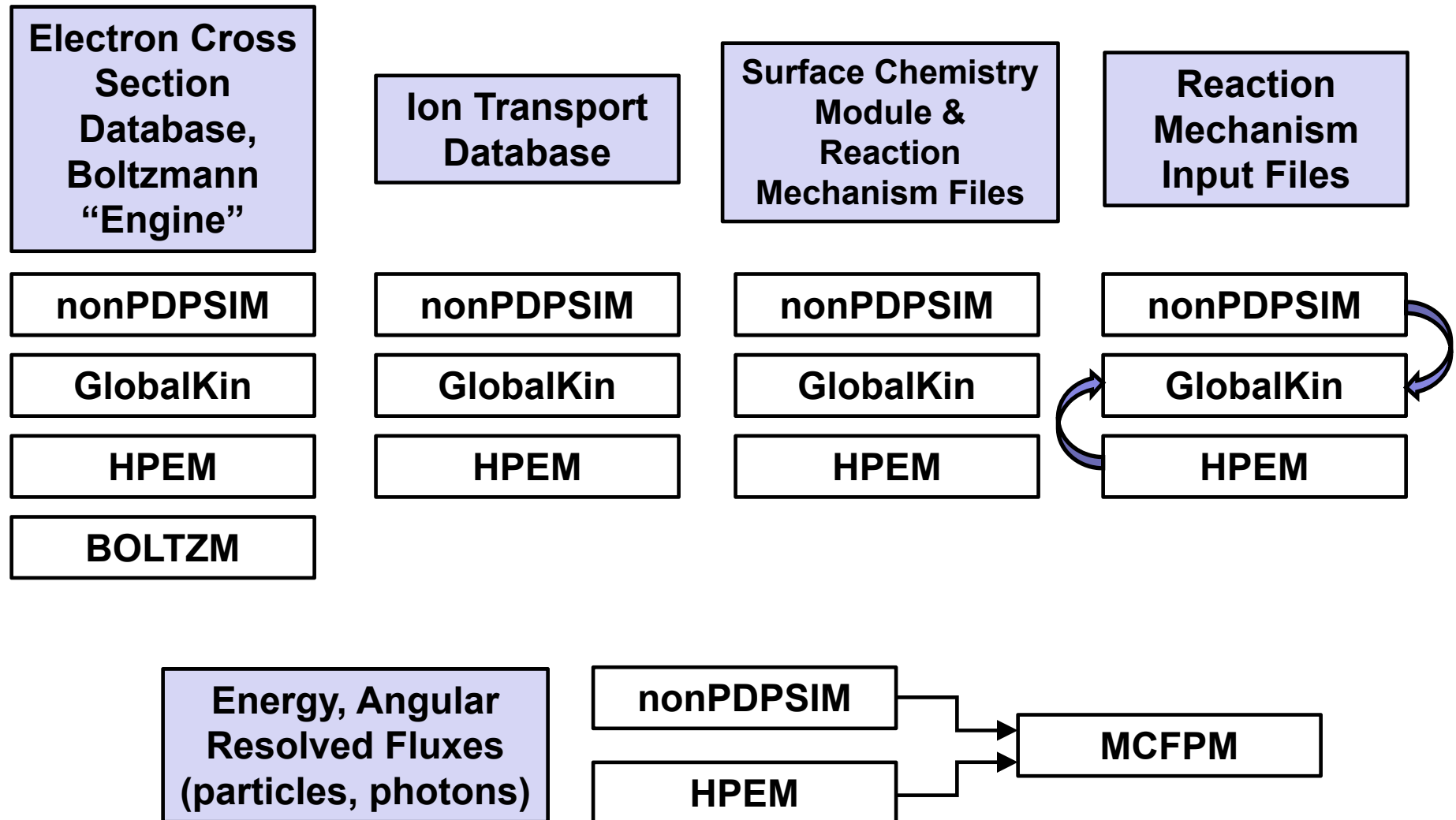
- In the absence of unlimited computing resources and data, decisions must be made on “precise” vs “conceptual” outcomes of models.



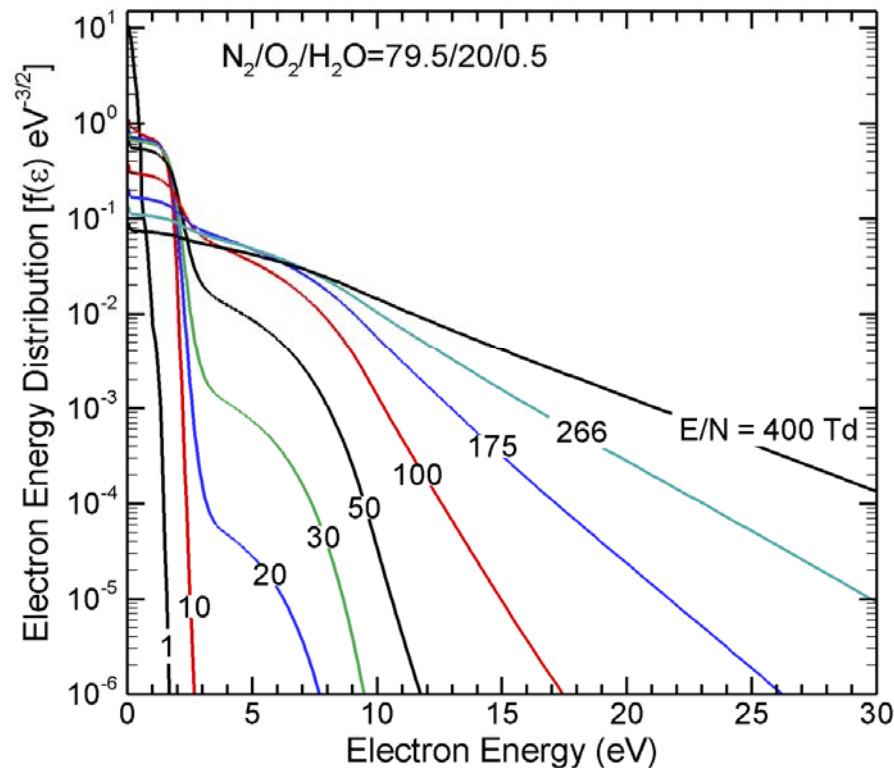
UM MODELING PORTFOLIO



COMMON COMPONENTS



UM MODELING PORTFOLIO: BOLTZM



- “Stand-alone” 2-term Spherical Harmonic Expansion solution of Boltzmann Equation
- Steady state, time dependent, beam-bulk, e-e, e-ion
- AC fields
- “Engine” is embedded in other codes

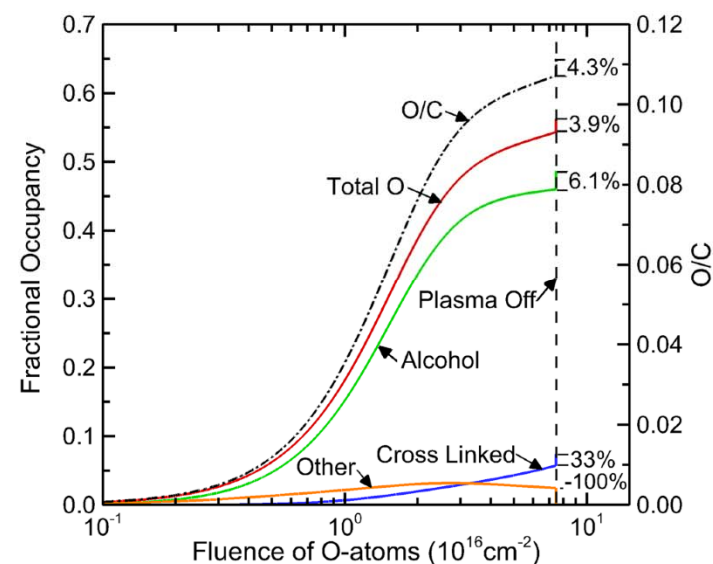
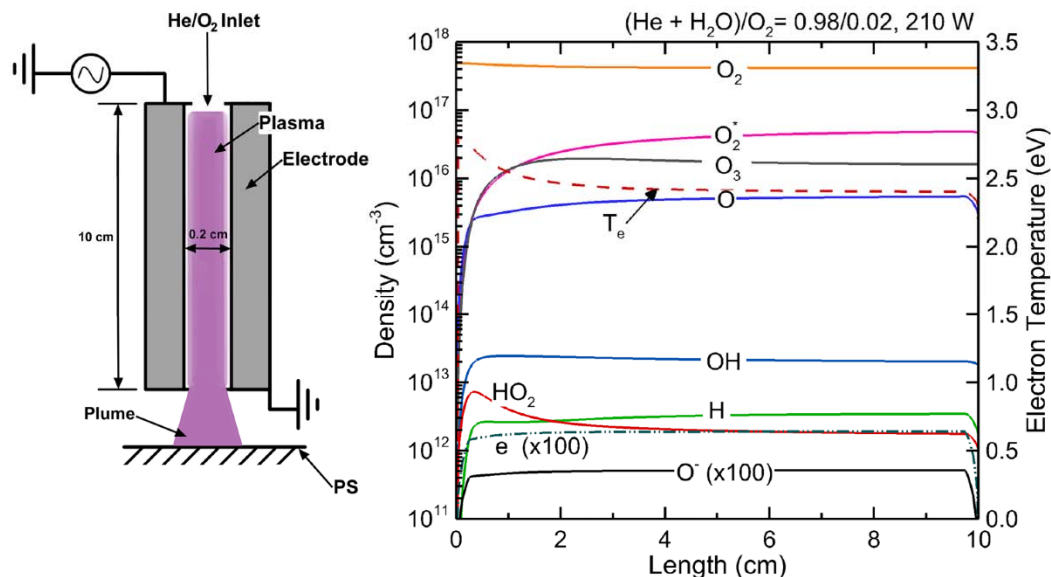
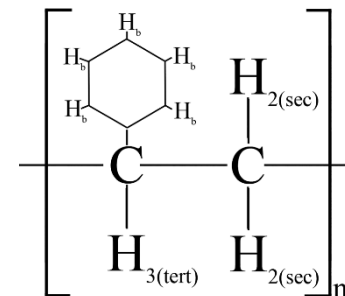
UM MODELING PORTFOLIO: GlobalKin

- **Global plasma chemistry and plasma surface interactions**
- **Surface chemistry**
- **Radiation transport**
- **Plug flow**
- **Boltzmann derived rate and transport coefficients**
- **Bulk and beam electrons**
- **Multiphase (dust particles, liquids)**
- **Surface Chemistry**

UM MODELING PORTFOLIO: GlobalKin

Atmospheric pressure plasma functionalization of polystyrene

Jordyn Polito, Mark Denning, Richard Stewart, David Frost, and Mark J. Kushner
J. Vac. Sci. Technol. A 40, 043001 (2022); doi: [10.1116/6.0001850](https://doi.org/10.1116/6.0001850)



- Plug flow model of He/H₂O/O₂ plasma jet into ambient humid air.
- Plume functionalizes a polystyrene polymer producing mostly alcohol groups.

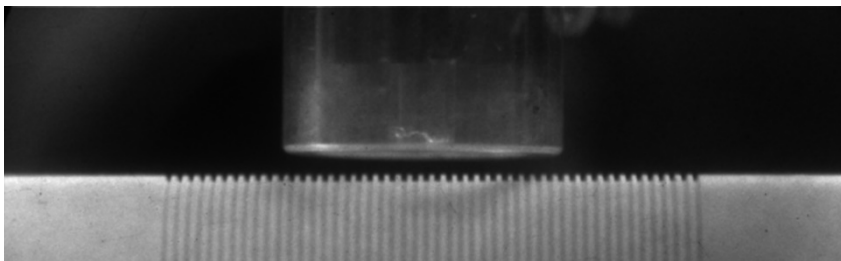
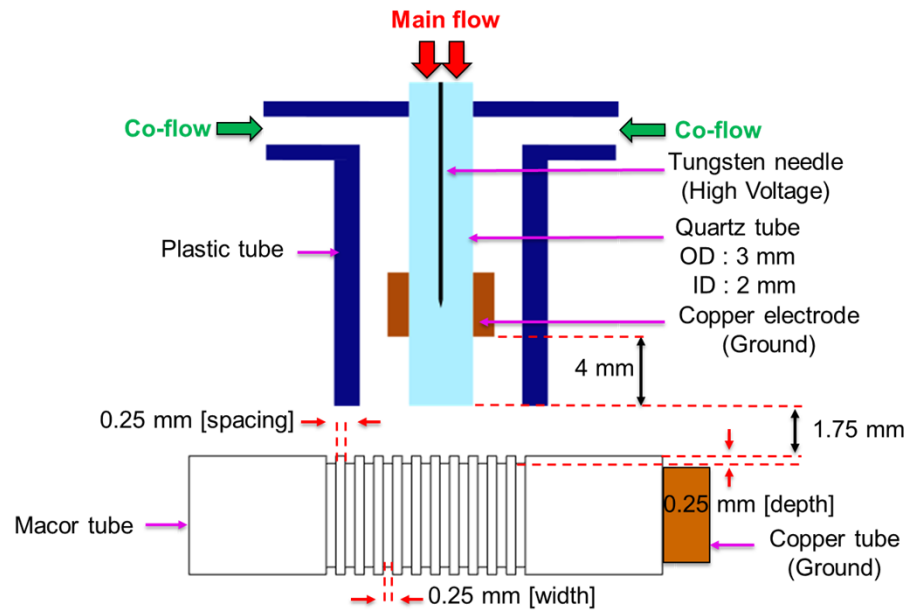
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UM MODELING PORTFOLIO: nonPDPSIM

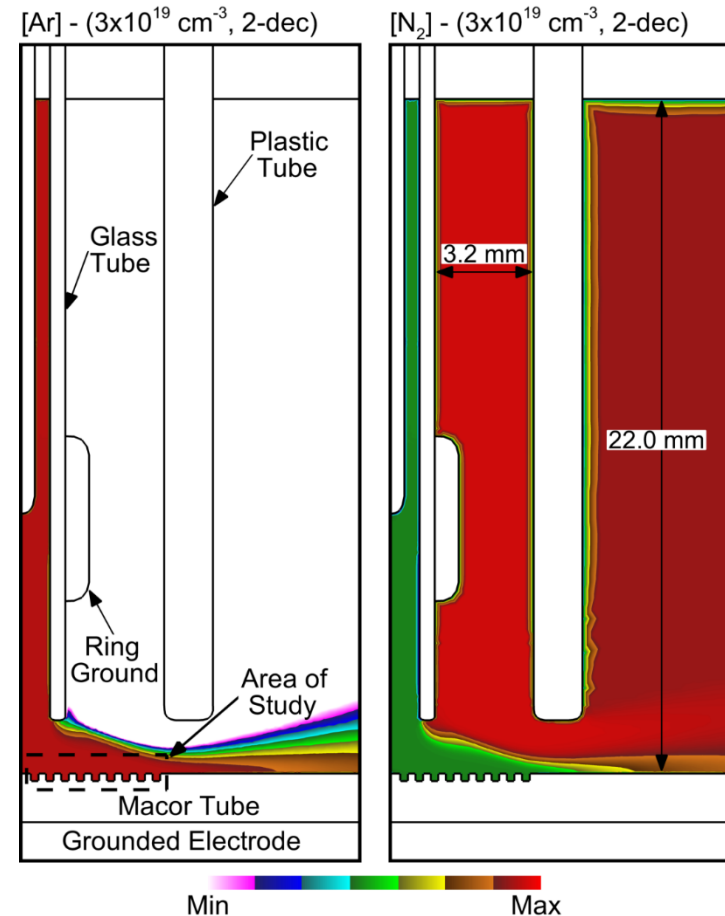
- **2-dimensional hybrid**
- **Multi-physics, multi-fluid**
- **Multiphase (plasma, liquids)**
- **Surface Chemistry**
- **Fluid foundation with kinetic electrons and ions for tracking distribution functions**
- **Radiation transport**
- **Electrostatic, magnetostatic, electromagnetic**
- **Tuned for higher pressures and liquids.**

Atmospheric pressure plasmas interacting with wet and dry microchannels: reverse surface ionization waves

Kseniia Konina, Sai Raskar, Igor V Adamovich and Mark J Kushner
 Plasma Sources Sci. Technol. **33** (2024) 015002



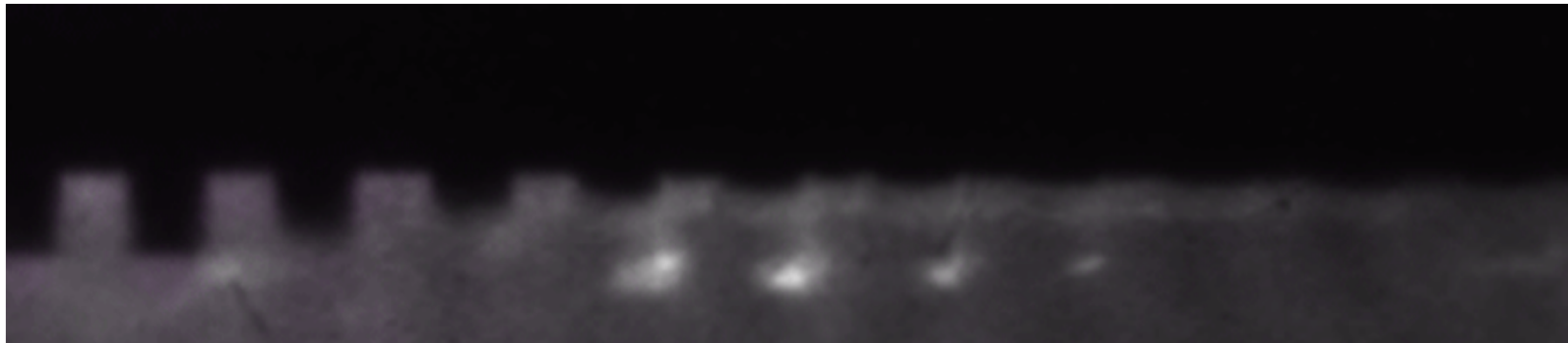
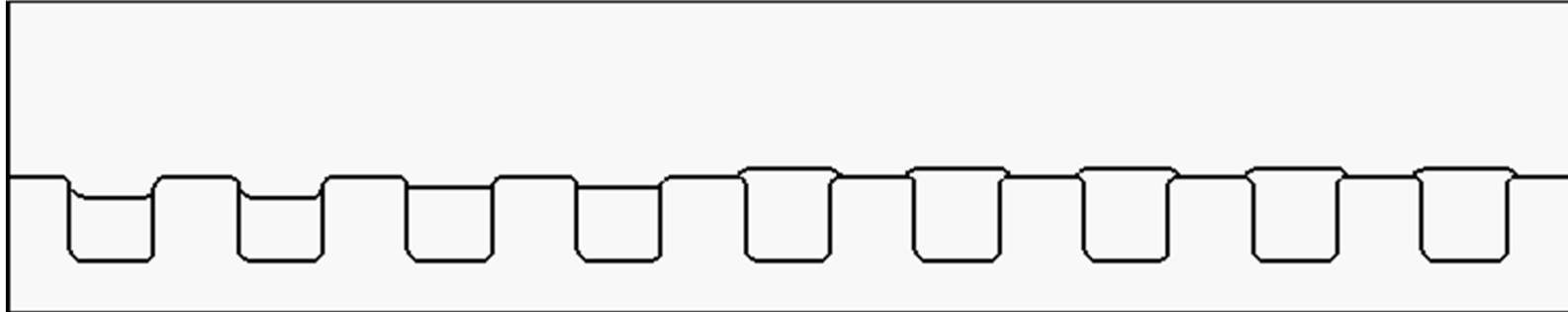
PLASMA JET ONTO WET MICROCHANNELS



SIW ON CHAIN OF WET CHANNELS

[e] ($1 \times 10^{14} \text{ cm}^{-3}$, 3-dec)

-25 kV, $\epsilon = 4$



- **Surface ionization wave in air propagating over water filled microchannels.**
- **Electric field enhance of convex water surfaces provide focusing of SIW and launching of reverse ionization waves**

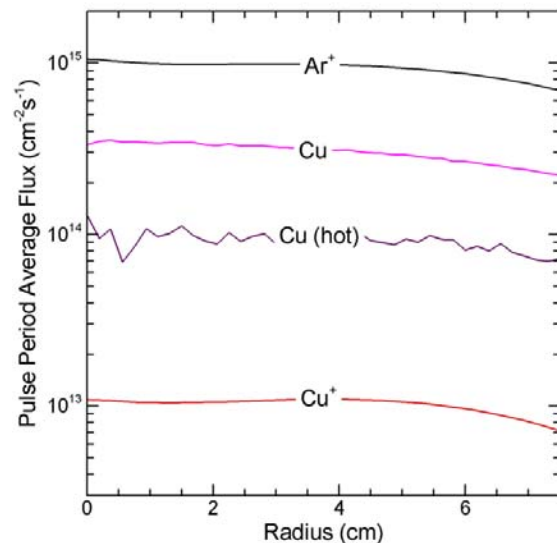
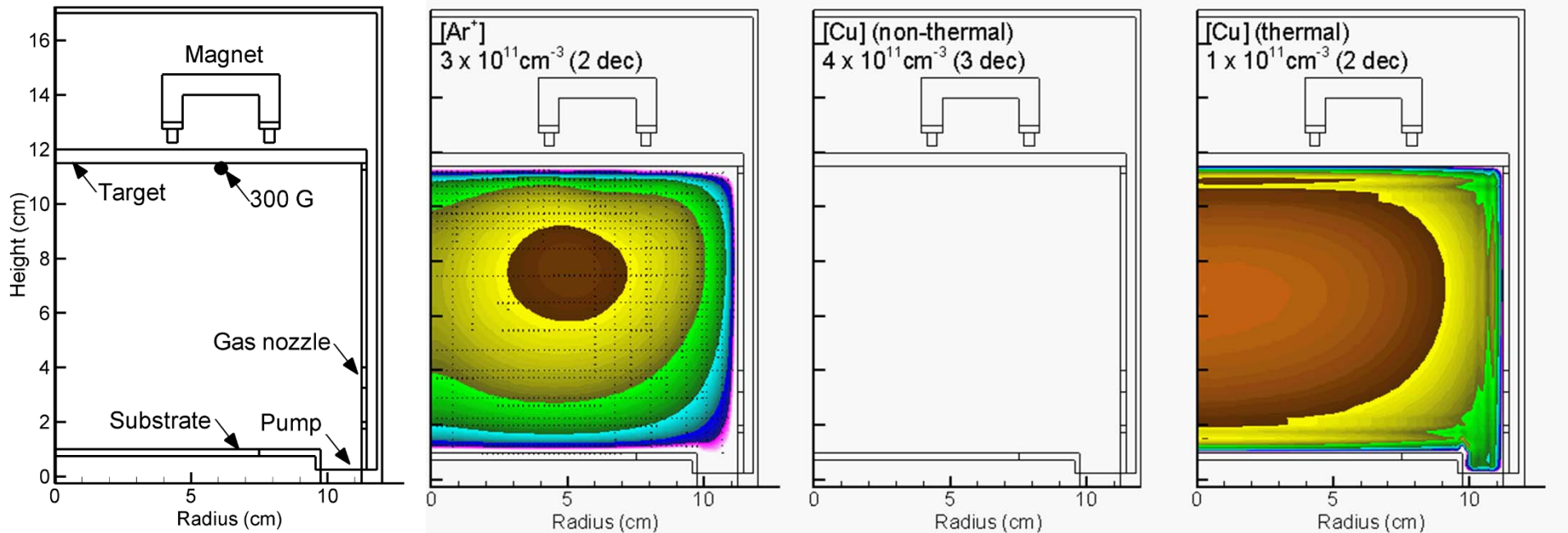
Animation Slide

UM MODELING PORTFOLIO: HPEM

HPEM (Hybrid Plasma Equipment Model)

- 2-dimensional, multi-physics, multi-fluid, hybrid
- Multiphase
- Surface chemistry
- Dust particle transport and growth
- Fluid foundation with kinetic electrons and ions for tracking distribution functions and generating rate/transport coefficients
- Radiation transport
- Sputtering
- Electrostatic, magnetostatic and electromagnetic
- ICP, CCP, beam, microwave, ECR, magnetron

HiPIMS-ION FLUXES AND SPUTTER SOURCES



- High power impulse magnetron sputtering (HiPIMS) – enables control by optimizing "overlap" of species incident onto substrate.
- HiPIMS-like pulsed PVD of Cu, Ar, 10 mTorr
- 8.33 kHz, 13.5 μs pulse (12% duty cycle), -400 V

Animation Slide

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MCFPM (Monte Carlo Feature Profile Model)

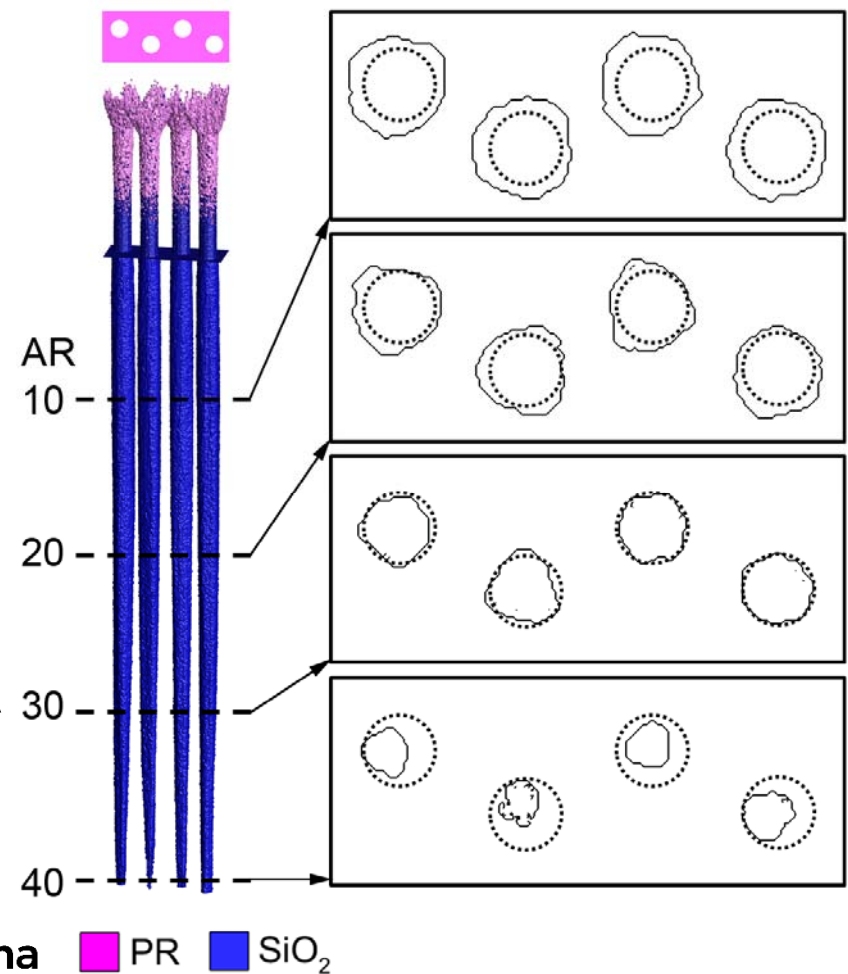
- Fluxes (energy, angle) onto surface from HPEM/nonPDPSIM
- 3-dimensional voxel-based feature profile evolution
- Surface and sub-surface reactions
- Radiation transport and photolytic reactions
- Adsorption, desorption, sputtering, passivation, implantation
- Charging
- Pattern distortion during plasma etching of SiO₂ High Aspect Ratio vias

Pattern dependent profile distortion during plasma etching of high aspect ratio features in SiO₂

Shuo Huang, Seungbo Shim, Sang Ki Nam, and Mark J. Kushner
J. Vac. Sci. Technol. A 38, 023001 (2020); doi: [10.1116/1.5132800](https://doi.org/10.1116/1.5132800)

ESCAMPIG_2024

UM MODELING PORTFOLIO: MCFPM



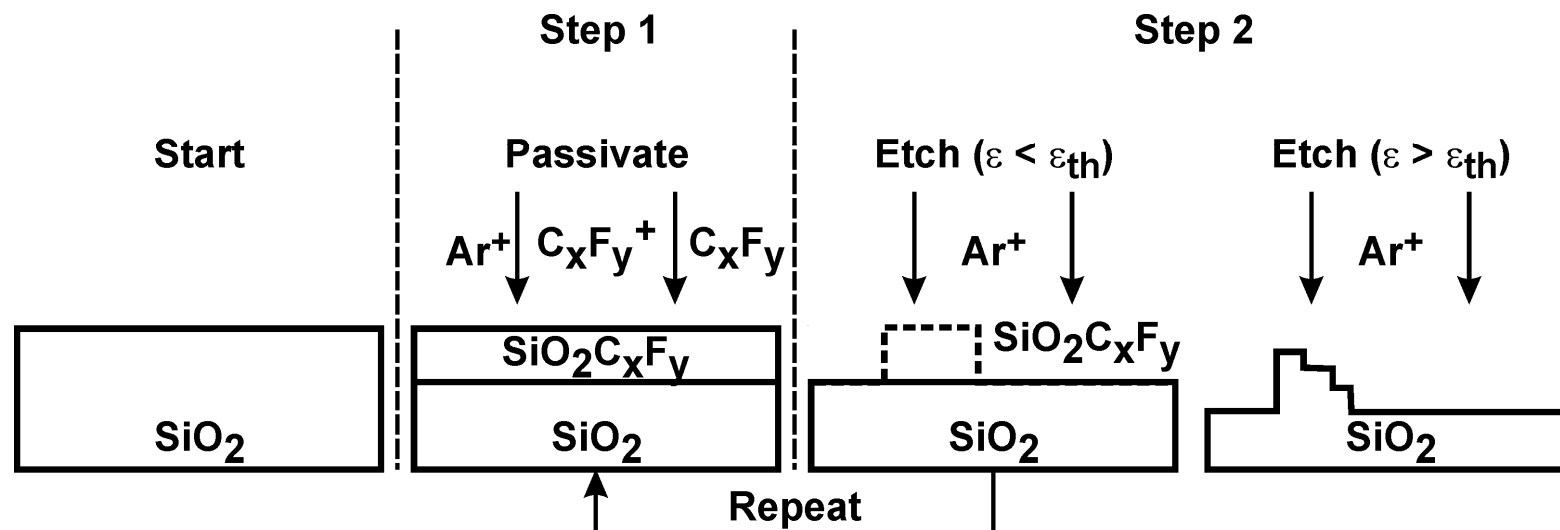
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ADAM ASKED...ASSESS IMPACT

- **Impact – well, that is a judgement call.**
- **Companies use the codes - plasma equipment, chip-manufacturers, material processing**
- **University and national laboratories use the codes.**
- **Hard to say why they use the codes(!) other than the codes address problems they care about and provide reasonable outcomes.**
- **Must be having some impact.**
- **That said, impact sometimes takes a while....**

PLASMA ATOMIC LAYER ETCHING

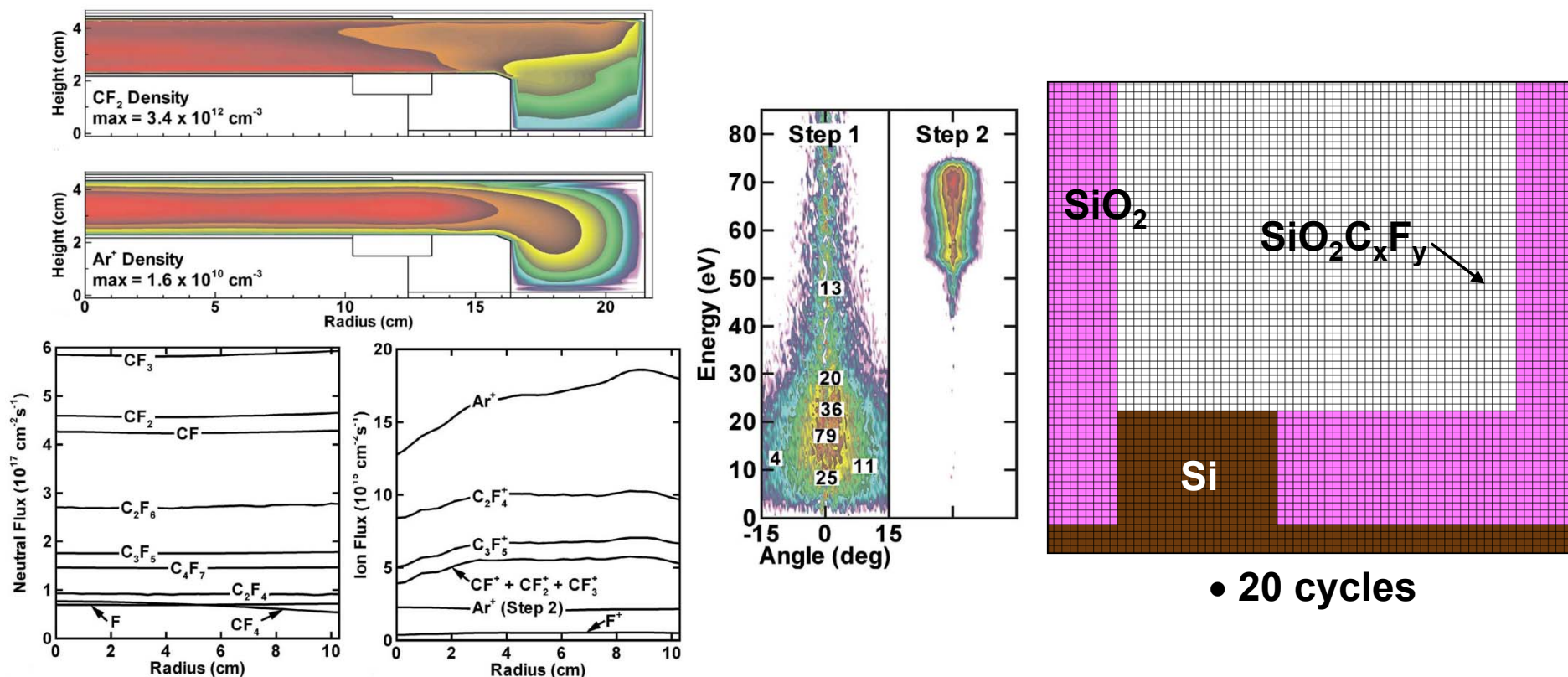
- Plasma atomic layer etching (PALE) – 2 step technique to remove a controlled amount of material for microelectronics fabrication.
- For dielectric materials (SiO_2 , Si_3N_4) PALE consists of deposition of a thin fluorocarbon polymer layer, followed by low energy ion bombardment.
- Controlled ion energy is important to prevent unwanted, non-selective physical sputtering.



Plasma atomic layer etching using conventional plasma equipment

Ankur Agarwal Mark J. Kushner

J. Vac. Sci. Technol. A 27(1), Jan/Feb 2009

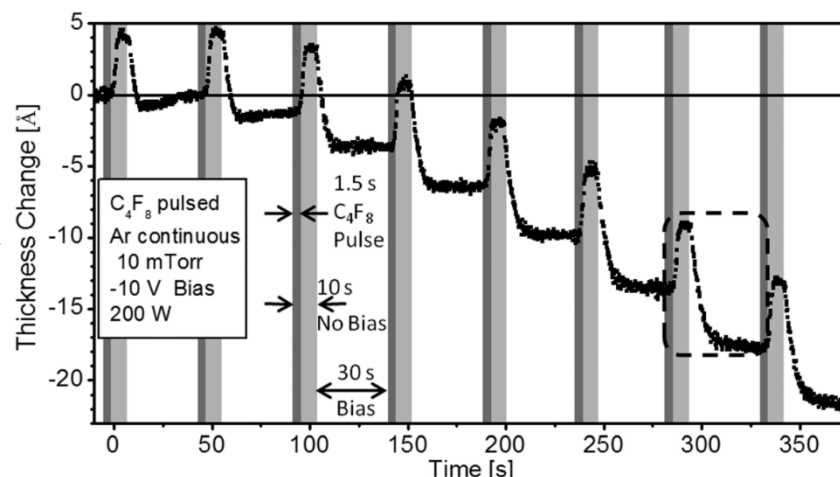
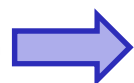
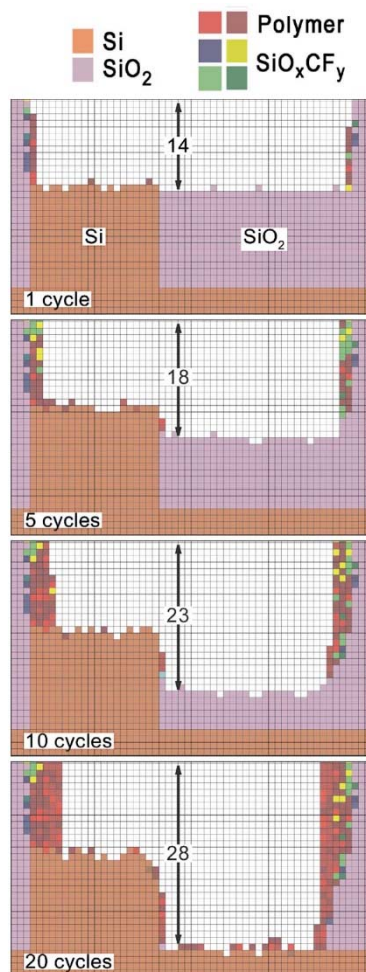


- Demonstrated concepts for selective PALE of SiO_2 using sequential pulsed capacitively coupled plasmas sustained in Ar/C_4F_8 and Ar with finely tuned ion energy control.

Animation Slide

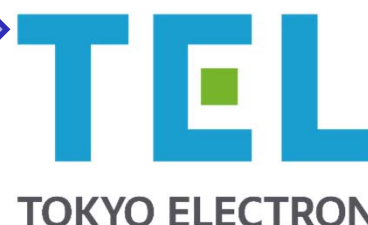
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Institute for Plasma Science & Engr.

PALE OF SiO₂ INTO HIGH VOLUME MANUFACTURING



Fluorocarbon assisted atomic layer etching of SiO₂ using cyclic Ar/C₄F₈ plasma
 Dominik Metzler, Robert L. Bruce, Sebastian Engelmann, Eric A. Joseph, and Gottlieb S. Oehrlein
 Journal of Vacuum Science & Technology A **32**, 020603 (2014)

• 2012-2014



• 2020-now?

Plasma atomic layer etching using conventional plasma equipment

Ankur Agarwal Mark J. Kushner

J. Vac. Sci. Technol. A **27**(1), Jan/Feb 2009

• 2007-2008

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CONCLUDING REMARKS

- Our focus in LTP modeling has been a multi-physics approach, attempting to address a broad range of technologies.
- Include fundamental processes while also being technology friendly.
- Lessons learned include:
 - Approach LTP modeling as a plasma physics problem first and a computer science problem second.
 - Enforce standards within your research group to enable several students to work on the same modeling platform and to enable codes to survive several generations of students.
 - Use models to explore new parameter spaces and concepts while accepting the risk that you may be shown not 100% correct years later.

POST-DOCTORAL RESEARCH POSITION

- **A Post-doctoral research position in computational low temperature plasmas and plasma surface interactions is available at the University of Michigan, USA.**
- **Contact: Prof. Mark J. Kushner, mjkush@umich.edu**